Process

To turn a 3-bit binary input into seven outputs, each corresponding to a part of representing said number in decimal, a decoder is required. Decoders are often used in circuitry to cover a wide variety of input combinations. In this specific lab, a decoder translates binary to the corresponding inputs of a 7-segment display.

In order to derive these outputs, a truth table can be used. To set up the truth table, set the inputs A, B, and C to each binary combination for numbers 0–7. These will then be compared with the inputs required to create each number on the display. In essence, this truth table will match the required combination of the 3 inputs to the desired 7 outputs needed to produce the desired number.

After creating a truth table, logic sets of minterms were gathered for each of the 7 segments. These minterm equations were then simplified using Karnaugh Maps to reduce them to their simplest form.

Using these equations, all that was left was to create the circuit. Using Logisim, each equation was implemented together to produce the intended circuit.

Difficulties

The most difficult part of this lab was the creation of the initial equations and the implementation of the simplified equations into Logisim. These difficulties stemmed more from a place of tedium than from a place of confusing logic. Understanding the general logic enabled me to focus on making sure I was righting equations properly and wiring the proper input to each logic gate.

Truth Table & Unsimplified Equations

Table1

Column A	Column B	Column C		A	B	C	D	E	F	G
0	0	0		1	1	1	1	1	1	0
0	0	1		0	1	1	0	0	0	0
0	1	0		1	1	0	1	1	0	1
0	1	1		1	1	1	1	0	0	1
1	0	0		0	1	1	0	0	1	1
1	0	1		1	0	1	1	0	1	1
1	1	0		1	0	1	1	1	1	1
1	1	1		1	1	1	0	0	0	0

A = (A'B'C')+(A'BC')+(A'BC)+(AB'C)+(ABC')+(ABC)

B = (A'B'C')+(A'B'C)+(A'BC')+(A'BC)+(AB'C')+(ABC)

C = (A'B'C')+(A'B'C)+(A'BC)+(AB'C')+(AB'C)+(ABC')+(ABC)

D = (A'B'C')+(A'BC')+(A'BC)+(AB'C)+(ABC')

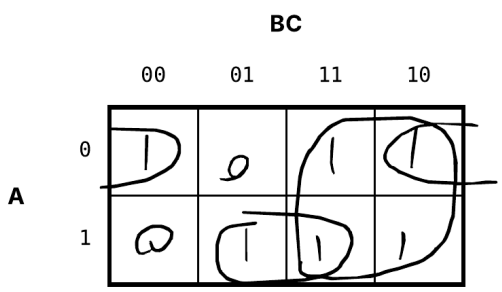
E = (A'B'C')+(A'BC')+(ABC')

F = (A'B'C')+(AB'C')+(AB'C)+(ABC')

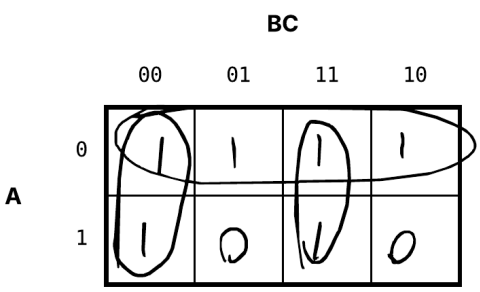
G = (A'BC')+(A'BC)+(AB'C')+(AB'C)+(ABC')

KMaps & Simplified Equations

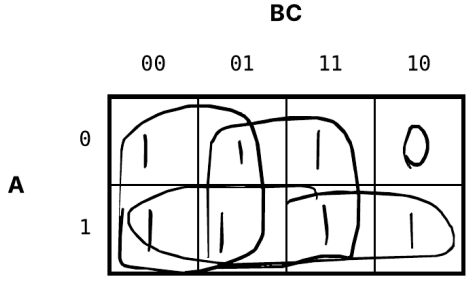
A



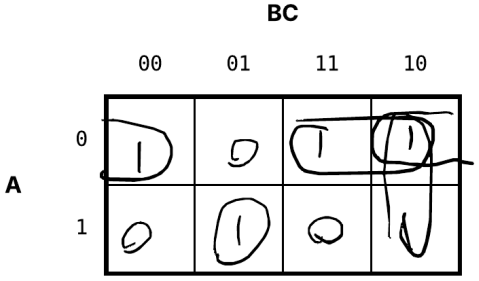
B



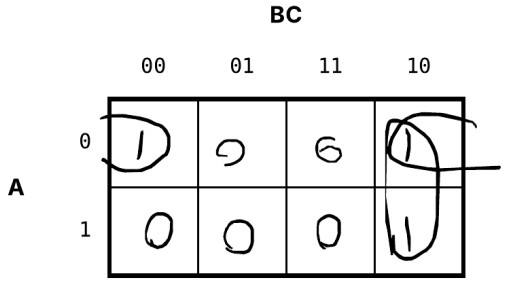
C



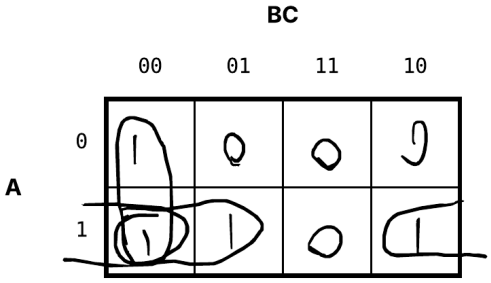
D



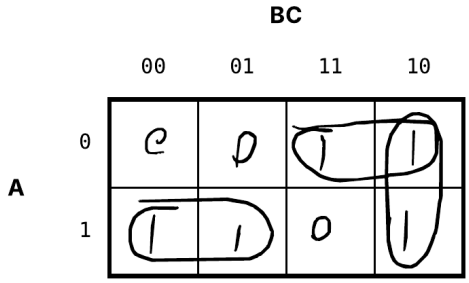
E



F



G



Final Equations:

$A = \bar{A}\bar{C} + B + AC$

$B = \bar{A} + \bar{B}\bar{C} + BC$

$C = A + \bar{B} + C$

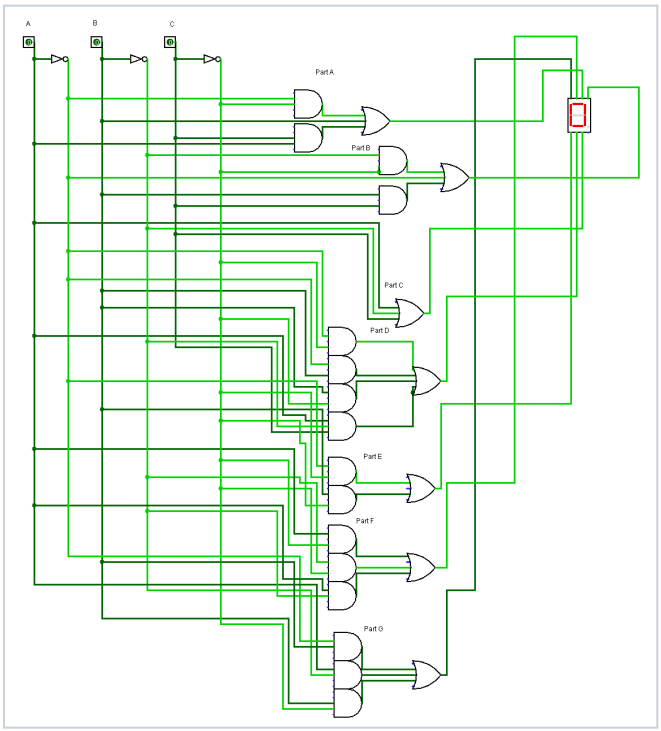
$D = \bar{A}\bar{C} + \bar{A}B + B\bar{C} + AB\bar{C}$

$E = \bar{A}\bar{C} + B\bar{C}$

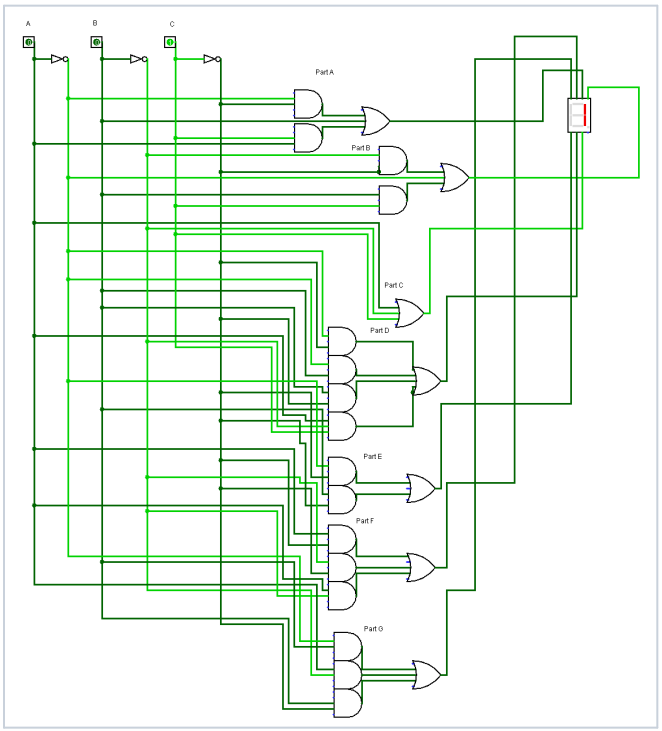
$F = A\bar{C} + \bar{B}\bar{C} + A\bar{B}$

$G = \bar{A}B + A\bar{B} + B\bar{C}$

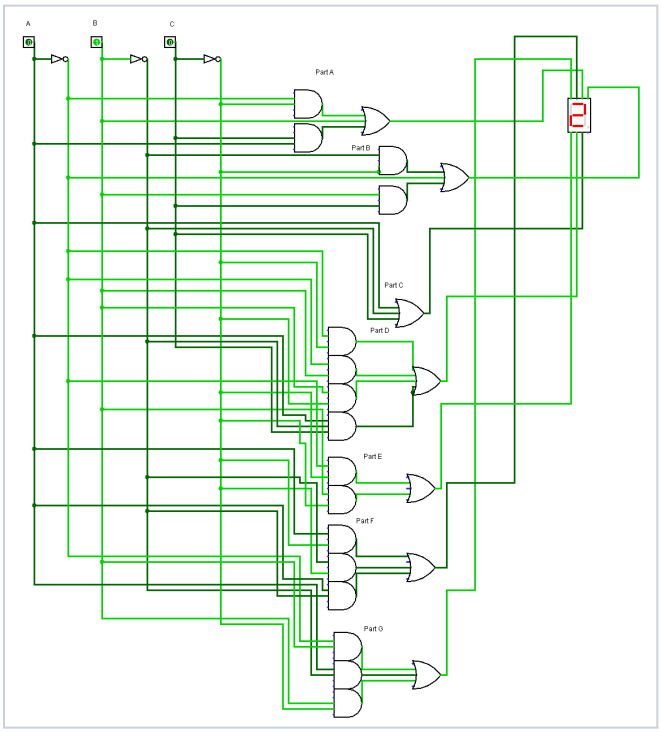
Logisim Models



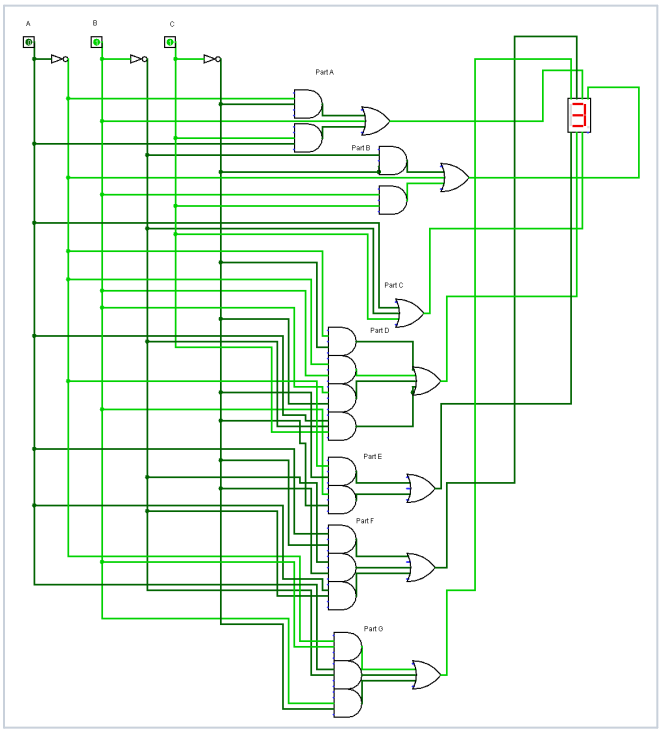
Logisim Model 0



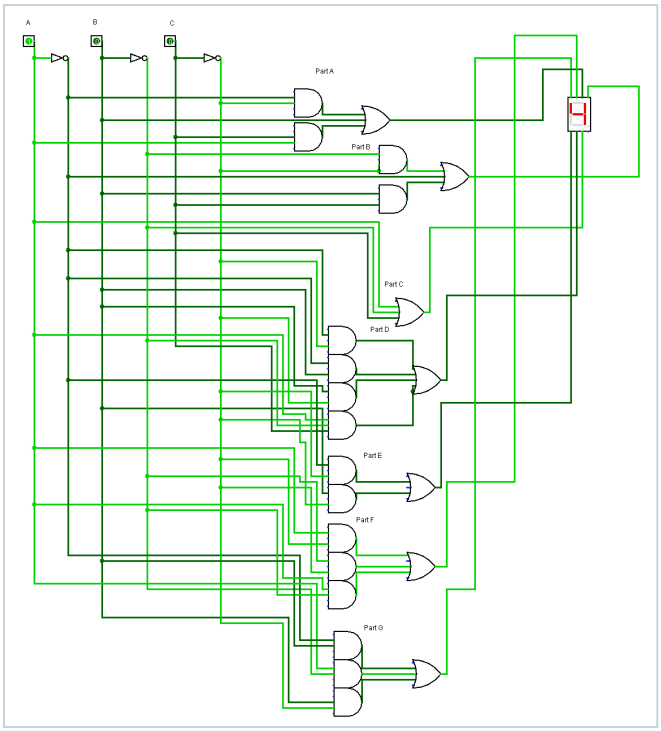
Logisim Model 1



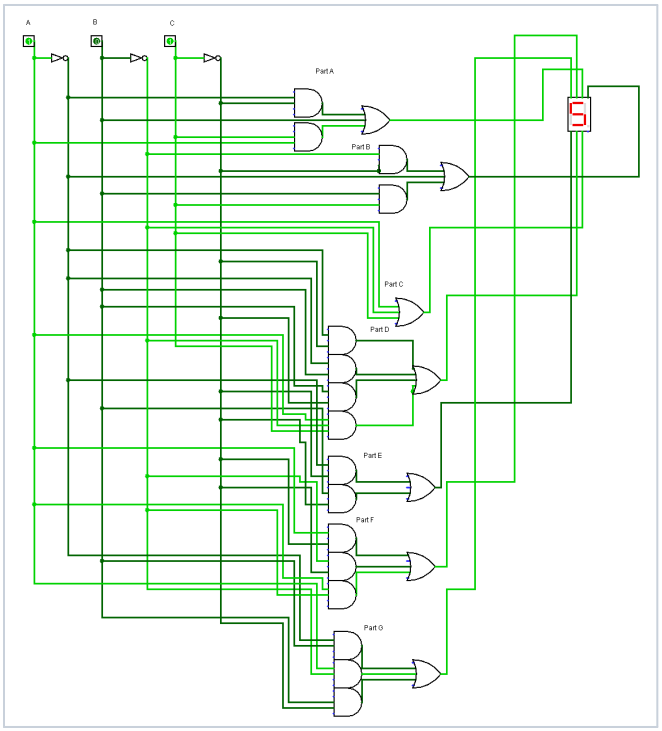
Logisim Model 2



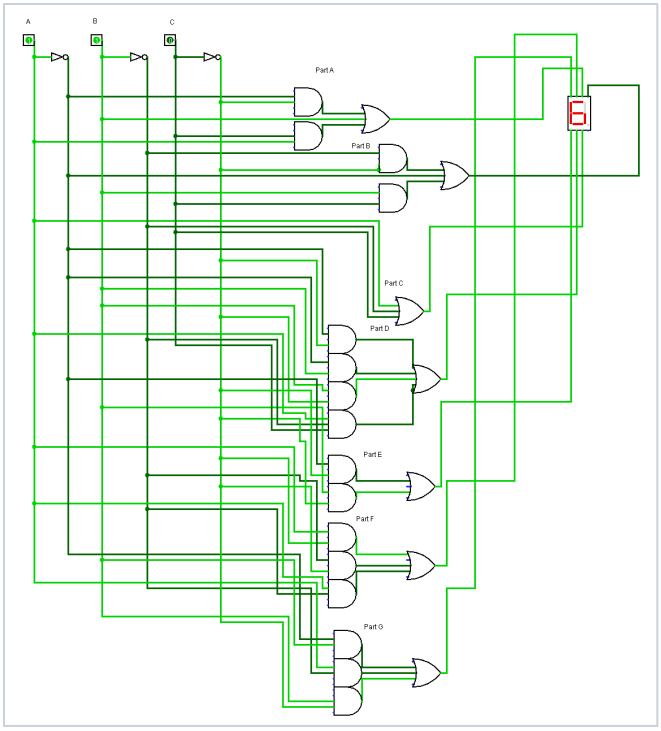
Logisim Model 3



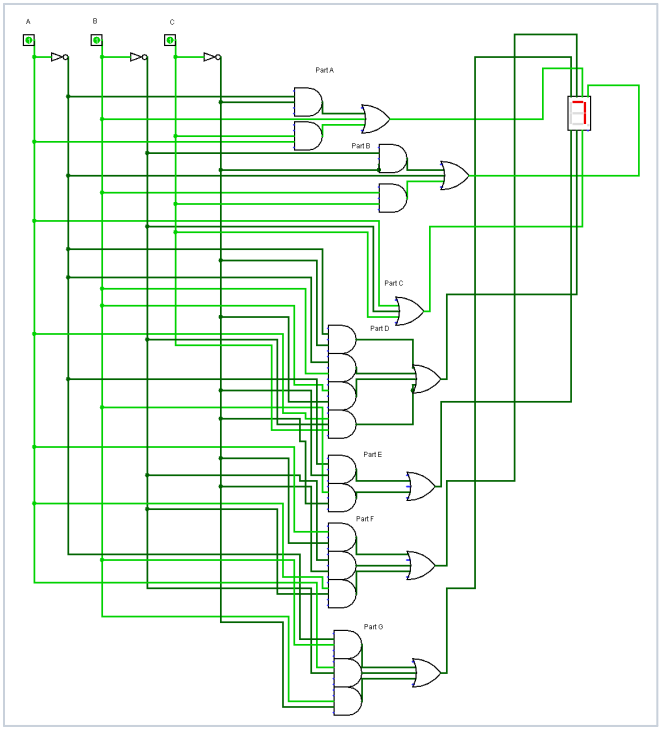
Logisim Model 4



Logisim Model 5



Logisim Model 6



Logisim Model 7